DARQ report

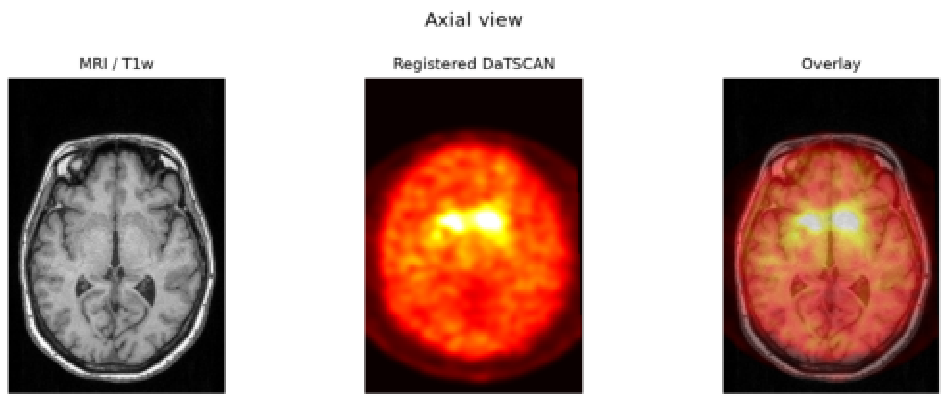
Subject: tutorial_subject

Report date: 2026-06-22 12:24

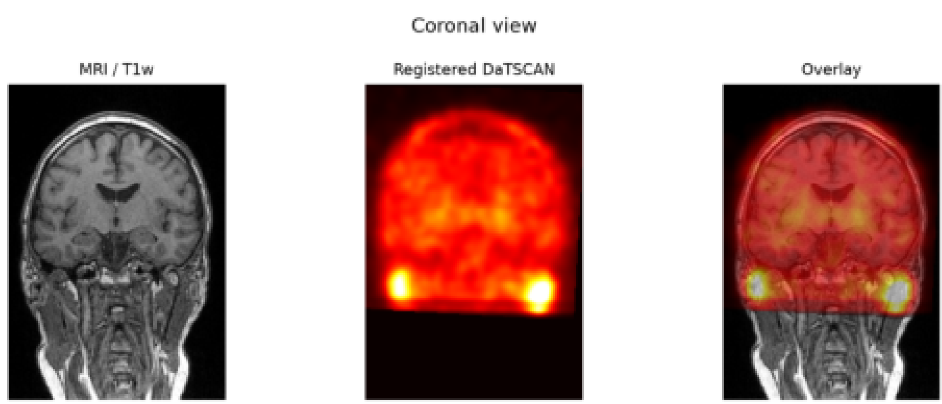
DaTSCAN quantification with MRI/T1w and SynthSeg

Main formulas: SBR = regional uptake / occipital uptake - 1; AI = 100 × (Right - Left) / mean(Right, Left).

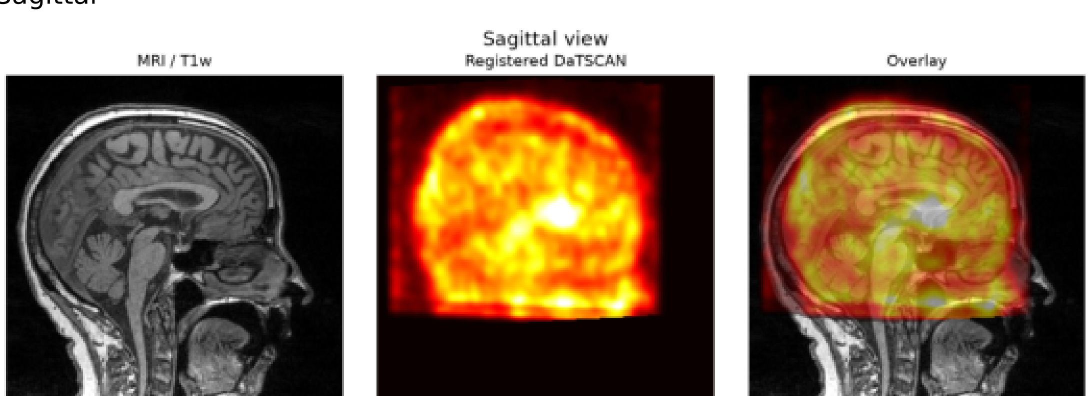
Axial



Coronal



Sagittal



Regional SBR summary

Region	Right SBR	Left SBR	Bilateral SBR	Asymmetry index (%)
Striatum	1.2125	1.3222	1.2669	-8.6589
Caudate	1.451	1.3499	1.4026	7.2229
Putamen	1.0379	1.3044	1.174	-22.7531

Hemisphere SBR table

Hemisphere	SBR striatum	SBR caudate	SBR putamen	Occipital reference	Registration loss
R	1.2125	1.451	1.0379	5734.8774	0.3191
L	1.3222	1.3499	1.3044	5633.6621	0.3191
Both	1.2669	1.4026	1.174	5685.0737	0.3191

Symmetry summary

Metric	Hemisphere	Caudate	Putamen
lncc	R	0.0056	0.0
lncc	L	0.3263	0.1791
lncc	Both	0.1615	0.0929
l2	R	2.6362	2.1919
l2	L	1.4446	1.1257
l2	Both	2.0573	1.6389

Execution log

```
Pipeline correctly executed.

Output folder:
C:\Users\Usuario\Desktop\Practicas_DaTSCAN\darq\gradio_runs\run_20260622_121430_tutorial_subject\output

SBR rows found: 21
Symmetry rows found: 30

Last terminal output:
ithm #
# ------------------------ #


Total sessions to process: N=1

Subject: 0  (0/1)
 * Preprocessing.
 * DaT symmetry and mask.
 * MRI DaT simulation and mask.
 * MRI-DaT registration.
    Optimizer: LBFGS lr=0.1  max_iter=10
    Losses: reg(w=1); reg_label(w=2); reg_uptake(w=0.0); reg_lr(w=0.5); regularizer(w=1)
    ###################
    # Training started #
    ###################


    * Epoch: 0
      Epoch summary: loss_reg: 0.355,w_loss_reg: 0.355,loss_reg_label: 0.001,w_loss_reg_label: 0.002,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.357,time_duration (s): 18.982


    * Epoch: 1
      Epoch summary: loss_reg: 0.346,w_loss_reg: 0.346,loss_reg_label: 0.001,w_loss_reg_label: 0.003,loss_reg_lr: 0.001,w_loss_reg_lr: 0.0,loss_regularizer: 0.002,w_loss_regularizer: 0.002,loss: 0.35,time_duration (s): 17.127


    * Epoch: 2
      Epoch summary: loss_reg: 0.339,w_loss_reg: 0.339,loss_reg_label: 0.002,w_loss_reg_label: 0.004,loss_reg_lr: 0.001,w_loss_reg_lr: 0.0,loss_regularizer: 0.001,w_loss_regularizer: 0.001,loss: 0.344,time_duration (s): 16.795


    * Epoch: 3
      Epoch summary: loss_reg: 0.312,w_loss_reg: 0.312,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.001,w_loss_reg_lr: 0.0,loss_regularizer: 0.001,w_loss_regularizer: 0.001,loss: 0.323,time_duration (s): 17.177


    * Epoch: 4
      Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.001,w_loss_regularizer: 0.001,loss: 0.32,time_duration (s): 17.001


    * Epoch: 5
      Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.009,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.319,time_duration (s): 17.036


    * Epoch: 6
      Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.319,time_duration (s): 18.48


    * Epoch: 7
      Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.319,time_duration (s): 18.057


    * Epoch: 8
      Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.319,time_duration (s): 16.695


    * Epoch: 9
      Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.319,time_duration (s): 16.516


    ####################
    # Training finished #
    ####################


    Final loss: 0.31907401164448074
Done in 531.82s.



Failed subjects: 0 / 1
  (none)

# ---------#
# All DONE #
# ---------#




Errores / warnings:
"clear" no se reconoce como un comando interno o externo,
programa o archivo por lotes ejecutable.
C:\Users\Usuario\Desktop\Practicas_DaTSCAN\darq\darq\processing.py:76: FutureWarning: `binary_opening` is deprecated since version 0.26 and will be removed in version 0.28. Use `skimage.morphology.opening` instead.
  mask = binary_opening(mask, ball(3))
C:\Users\Usuario\Desktop\Practicas_DaTSCAN\darq\darq\processing.py:340: FutureWarning: `binary_dilation` is deprecated since version 0.26 and will be removed in version 0.28. Use `skimage.morphology.dilation` instead. Note the lack of mirroring for non-symmetric footprints (see docstring notes).
  mask_dilated = binary_dilation(mask_symm, np.ones((10, 10, 10)))
C:\Users\Usuario\Desktop\Practicas_DaTSCAN\darq\darq\src\preprocessing.py:63: FutureWarning: `binary_opening` is deprecated since version 0.26 and will be removed in version 0.28. Use `skimage.morphology.opening` instead.
  image = binary_opening(image, self.struct_fn(data_dict[k]))
```